QuickDraw — trained bases vs classical baselines

1. Main result

We introduce a parametric quantum-circuit family of unitary image bases with $O(N \log N)$ forward / inverse cost and $O(\log N)$ trainable parameters per dimension. **(i)** On line-drawing imagery (QuickDraw, 32×32), the trained block-wrapped basis exceeds every linear classical baseline tested — including the dataset-fitted optimum BlockPCA-8 — by **+5.97 dB** PSNR over BlockDCT-8 at $\rho = 0.20$, after ≈ 5 minutes of training on a single RTX 3090. **(ii)** On natural images (DIV2K, 256×256), the same

family matches BlockDCT-8 to within ≈ 0.3 dB, saturating the **Ahmed–Natarajan–Rao limit** under which the KLT of a stationary Gaussian AR(1) source converges to the DCT. **(iii)** Accordingly, the parametric basis offers a rigorous improvement over BlockDCT on image distributions whose patch statistics depart from stationary Gaussian AR(1), and parity on those that do not.

2. Setup + pipeline

$$m = n = 5, \quad d = 2^m \cdot 2^n = 32 \times 32 = 1024$$

$$N_{\text{train}} = 500, \quad N_{\text{test}} = 50, \quad \text{seed} = 42, \quad \rho \in \{0.05, 0.10, 0.15, 0.20\}$$

Per image $x \in [0,1]^d$ with basis Φ:

$$y = \Phi x \quad \text{(forward)}$$

$$k = \lfloor \rho \cdot d \rfloor, \quad y_i' = \begin{cases} y_i & \text{if } |y_i| \in \text{top-}k \\ 0 & \text{else} \end{cases} \quad \text{(compression)}$$

$$\hat{x} = \text{clip}(\Phi^{-1} y', 0, 1), \quad \text{MSE} = \frac{1}{d}\|x - \hat{x}\|_2^2, \quad \text{PSNR} = 10 \log_{10}(1/\text{MSE})$$

For block transforms: same pipeline applied independently to each 8×8 tile, top-k pooled across all blocks.

AR(1) (first-order autoregressive). Each pixel = $\rho_{\text{AR}} \times$ previous + small Gaussian noise:

$$x_n = \rho_{\text{AR}} x_{n-1} + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

$\rho_{\text{AR}} \to 1$ Gaussian $\to$ KLT collapses to DCT (Ahmed–Natarajan–Rao). Empirical lag-1 autocorrelation $\hat{\rho}_{\text{AR}} = \frac{1}{2}(\hat{\rho}_{\text{row}} + \hat{\rho}_{\text{col}})$ across multiple samples: **DIV2K natural images** (4 different scenes, centre 256×256 crops) cluster in $\hat{\rho}_{\text{AR}} \in [0.84, 0.99]$ — close to the image-row-like regime, $\rho \approx 1$, where BlockDCT is near-optimal; **QuickDraw drawings** (3 samples, single 28×28 each) cluster tightly at $\hat{\rho}_{\text{AR}} \approx 0.70$ — further from the AR(1)–Gaussian limit, leaving room for a trained basis to beat BlockDCT. The two distributions barely overlap, exactly the QuickDraw vs DIV2K asymmetry the results table shows. Distinct from the keep-ratio ρ above.

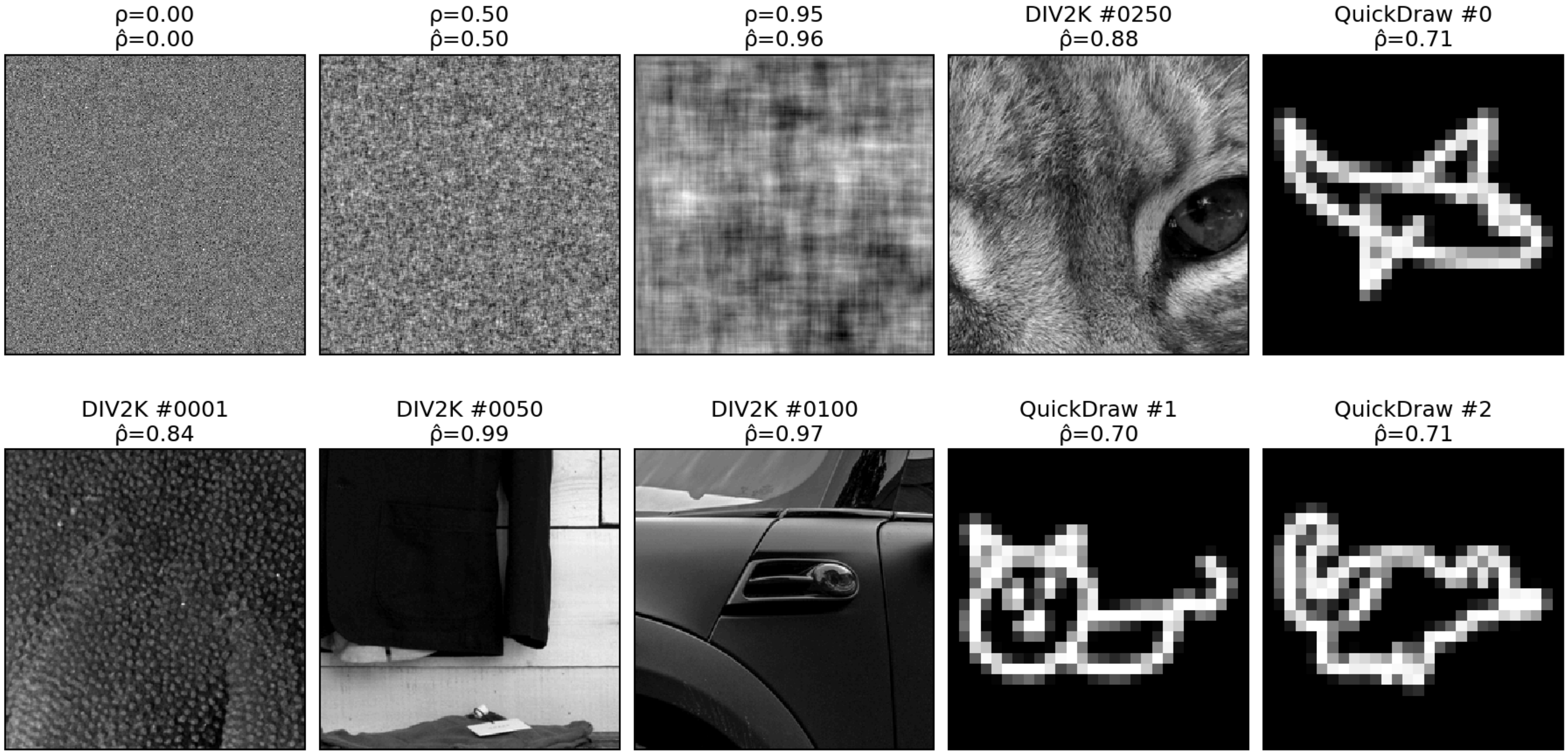


Figure 1: **Top row**: three synthetic AR(1)–Gaussian fields at $\rho = 0, 0.5, 0.95$ (sanity-checking $\hat{\rho}$); DIV2K 250 centre crop; QuickDraw drawing 0 **Bottom row**: three more DIV2K samples (1, 50, 100) and two more QuickDraw drawings (1, 2) — confirming the $\hat{\rho}_{\text{AR}}$ values are robust across samples and the two datasets cluster at different points on the AR(1) axis. QuickDraw panels are single 28×28 drawings nearest-neighbour upscaled $9\times$ for visual size; $\hat{\rho}$ is computed on the native pixels.

3. Fits — what is Φ for each method?

method	Φ source	Φ shape	forward $y =$	compression $y' =$	inverse $x =$
`bd_pca` (bilateral 2D-PCA)	U, V from SVDs of column- and row-stacked centered images: $\Sigma_{\text{col}} = \frac{1}{NW-1}\sum_i (X_i - \vert(X))(X_i - \vert(X))^T$ $(H \times H)$ $\Sigma_{\text{row}} = \frac{1}{NH-1}\sum_i (X_i - \vert(X))^T(X_i - \vert(X))$ $(W \times W)$ fit on $N = 500$ images at $H = W = 32$ ($NH = NW = 16000$ samples per axis)	U is $H \times H = 32 \times 32$, full rank V is $W \times W = 32 \times 32$, full rank (each axis has $16000 \gg 32$ samples)	$Y = U^T(X - \vert(X))V$, shape $H \times W$	top-k on Y: $Y_{ij} \cdot \mathbb{1}[\![\lvert Y_{ij}\rvert \geq \tau_k]\!]$, $k = \lfloor \rho \cdot HW \rfloor = \lfloor \rho \cdot d \rfloor$ (same rate as DCT)	$UY'V^T + \vert(X)$
`block_pca_8`	$\Phi = V_b^T$, where $\Sigma_b = V_b \Lambda_b V_b^T$, $\Sigma_b = \frac{1}{N_p-1}\sum_{j=1}^{N_p}(p_j - \mu_b)(p_j - \mu_b)^T$ fit on $N_p = 8000$ patches	Φ is 64×64 per block, full rank (Σ_b is $d_b \times d_b = 64 \times 64$ regardless of N_p) shared across all 4×4 blocks	per block: $\Phi(p - \mu_b)$	top-k pooled across all blocks, keep $k = \lfloor \rho \cdot 1024 \rfloor$ largest (rank: per-block keep $i = 0, ..., k_b - 1$, $k_b = \lfloor \rho \cdot 64 \rfloor$)	per block: $\Phi^T y' + \mu_b$, then re-tile
`dct` (global)	closed-form (DCT-II): $\Phi[k, n] = \alpha_k \cos(\pi(n + 1/2)k/N)$ $\alpha_0 = \sqrt{\frac{1}{N}}$, $\alpha_{k\geq 1} = \sqrt{\frac{2}{N}}$ $N = 32$, no fit, no mean	$d \times d$ analytically	Φx (no mean)	top-k: $y_i \cdot \mathbb{1}[\![\lvert y_i\rvert \geq \tau_k]\!]$ (rank: keep $i = 0, ..., k - 1$)	$\Phi^T y'$
`block_dct_8`	closed-form (DCT-II) at $N = 8$: $\Phi[k, n] = \alpha_k \cos(\pi(n + 1/2)k/8)$ shared across all blocks	64×64 per block analytically tiled identically across the image	per block: Φp	top-k pooled across all blocks (rank: per-block keep $i = 0, ..., k_b - 1$, $k_b = \lfloor \rho \cdot 64 \rfloor$)	per block: $\Phi^T y'$, then re-tile

Table 1: Concrete definition of Φ and the full pipeline (forward → compression → inverse) for each method. All four Φ are **orthogonal** ($\Phi\Phi^T = I$), so inverse = transpose. 2D = separable (apply 1D transform along rows then columns).

Top-k rule (default, headline results): keep the k coefficients with largest **|magnitude|**, set the rest to zero — pooled across all blocks for block transforms. **Rank rule** (in parentheses; used by the `_rank` control variants): keep the **first** k coefficients in the basis's natural ordering. For **PCA** this means the k **highest-eigenvalue** directions (rows of V^T are sorted by descending λ); for **DCT** this means the k **lowest-frequency** components (DC first, then increasing frequency $k = 0, 1, 2, ...$). See the Results table for the 4 to 9 dB cost of switching rule on the same fitted basis.

3.1. Why DCT $\approx$ block PCA in the limit

$$\Phi_{\text{block\_pca\_8}} = \mathrm{KLT}(\Sigma_b) \quad \longrightarrow \quad \Phi_{\text{block\_dct\_8}} = \mathrm{KLT}\left(\lim_{\rho\to 1} \Sigma_{\mathrm{AR}(1)}(\rho)\right)$$

DCT = what KLT becomes if the patch covariance is the analytic AR(1)–Gaussian limit. Empirically the two differ by only 0.68 dB on QuickDraw — natural patches are nearly AR(1)–Gaussian.

4. Results — PSNR (dB), seed=42

unblocked (full 32×32)	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
Trained PDFT bases (ours)				
⋆ `qft`	**16.72**	**19.58**	**22.06**	**24.35**
⋆ `entangled_qft`	16.72	19.58	22.05	24.35
⋆ `tebd`	16.64	19.40	21.79	24.01
Classical, top-k rule				
`bd_pca`	**18.63**	**21.99**	**24.87**	**27.57**
`dct`	16.13	18.44	20.33	22.03
`fft`	15.26	17.32	19.01	20.56
Classical, rank rule (control)				
`dct_rank`	13.84	15.24	16.49	17.63

8×8 block-wrapped	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
Trained PDFT bases (ours)				
⋆ `rich`	**18.82**	**23.71**	**28.15**	**32.60**
⋆ `real_rich`	18.79	23.68	28.05	32.37
⋆ `blocked`	18.12	22.41	26.20	30.06
Classical, top-k rule				
`block_dct_8`	**17.20**	**20.70**	**23.72**	**26.63**
`block_pca_8`	16.95	20.46	23.34	25.95
`block_fft_8`	15.97	18.74	21.09	23.34
Classical, rank rule (control)				
`block_pca_8_rank`	**13.60**	**14.95**	**16.12**	**17.16**
`block_dct_8_rank`	13.50	14.81	15.78	16.97

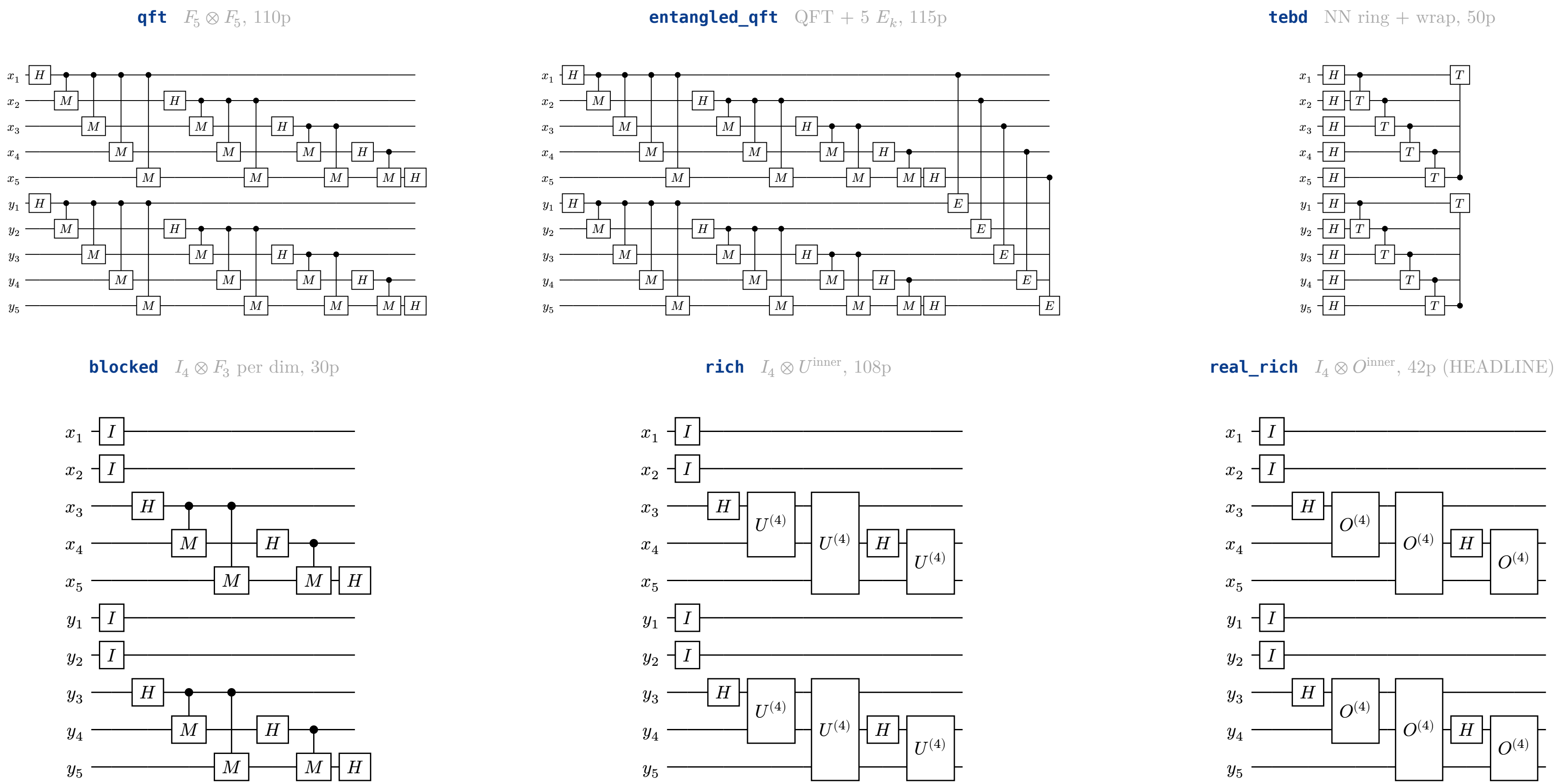
Table 2: Side-by-side: unblocked / full-image methods (left, gray header) vs 8×8 block-wrapped methods (right, light blue header). Each column groups trained PDFT bases, classical top-k baselines, and the rank-rule control. ⋆ = trained PDFT basis (this work); unmarked rows

5. classical baselines. Bold = best-in-group at each

keep ratio. The overall headline winner is `rich` ⋆ (block-wrapped, all four ratios). `mera` omitted (incompatible at $m + n = 10$).

5.1. Tensor networks of the trained bases at $m = n = 5$

10-qubit layout (rows $x_1, .., x_5$ top, cols $y_1, .., y_5$ bottom). Seven sub-circuits arranged in a 4-column grid for side-by-side comparison. All separable except `entangled_qft`.



Listing 1: Six trained-basis topologies at QuickDraw geometry ($m = n = 5$, 10 qubits per circuit). Top row — unblocked: `qft`, `entangled_qft`, `tebd` (all at the same scale and row-spacing so their 10-wire heights match exactly). Bottom row — block-wrapped: `blocked`, `rich`, `real_rich` (headline). `mera` is structurally inapplicable here ($m + n = 10 \neq 2^p$) and omitted. Param counts cover both row and col registers.

5.2. Geometry comparison at a glance

basis	**topology**	**row–col coupling**	**params/dim**	**PSNR at $\rho = 0.20$**
`qft`	all-to-all per dim	none	55	24.35
`entangled_qft`	all-to-all + 5 cross	yes (E_k)	60	24.35
`tebd`	NN ring per dim	none	25	24.01
`blocked`	outer-I + inner QFT-3	none	15	30.06
`rich`	outer-I + inner $U^{(4)}$	none	54	32.60
`real_rich`	outer-I + inner $O^{(4)}$	none	21	32.37

Table 3: All bases at QuickDraw geometry $m = n = 5$. Param counts are per dim (multiply by 2 for full transform unless row/col are tied). PSNR from Table earlier in this doc.

6. Reconstructions — 2 representative images × keep ratios × bases

6.1. Image 0 — cat sketch

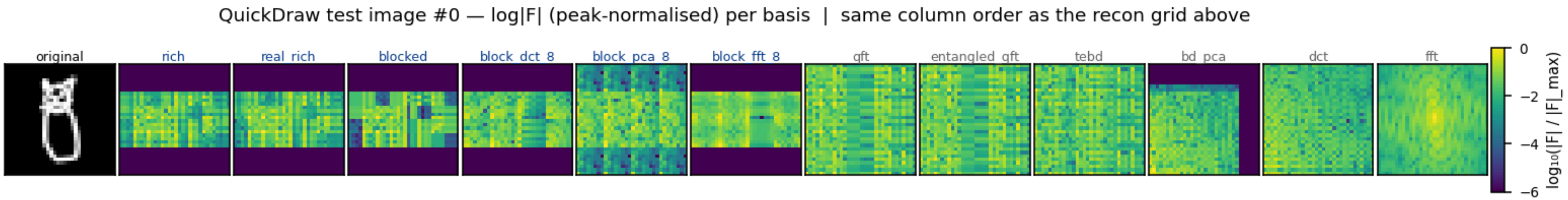


Figure 2: Frequency-space spectra (peak-normalised log|F|, viridis, shared color scale) for image 0 under each basis. Same column order as the recon grid below. Note the trained block-wrapped bases (blue headers) push energy into a small number of low-coefficient cells per block — visible as the bright clusters in `rich`/`real_rich`/`blocked` and the band structure in `block_dct_8`/`block_pca_8`. The `bd_pca` panel shows energy concentrated in the top-left corner (low-frequency / high-eigenvalue) along both axes — the separable-KLT analog of the DCT spectrum.

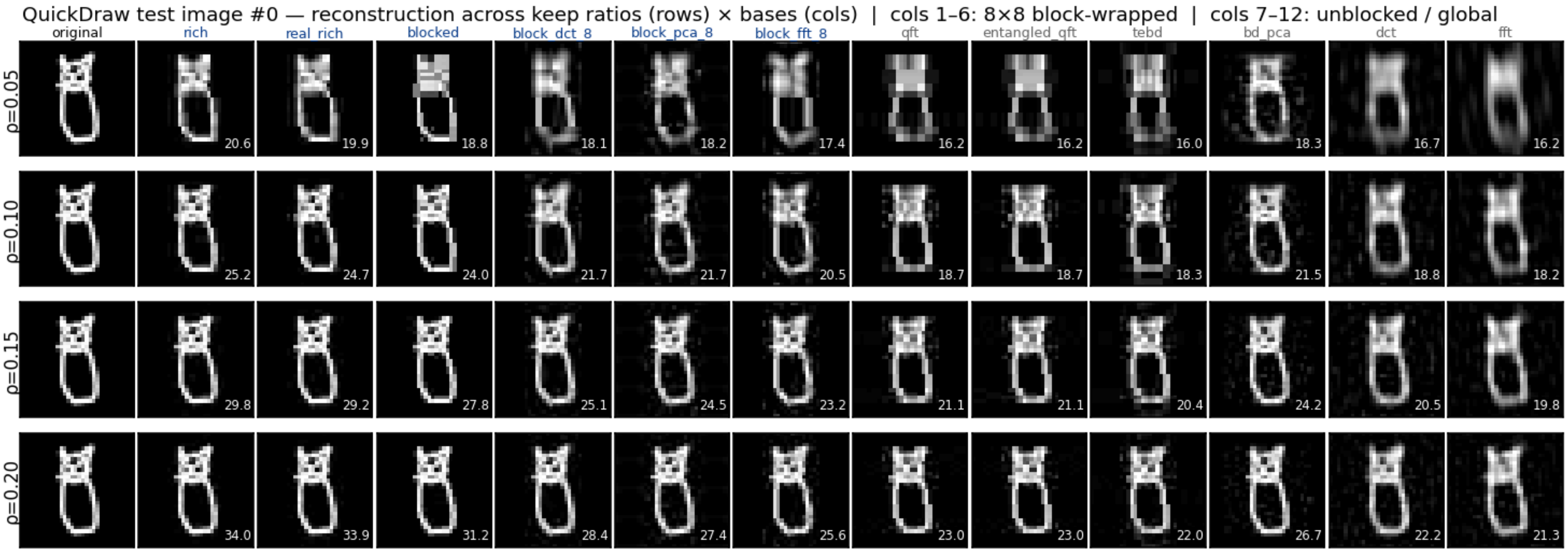


Figure 3: QuickDraw test image 0 reconstructed at four keep ratios $\rho \in \{0.05, 0.10, 0.15, 0.20\}$ (rows). Same column order as the freq-space figure above. PSNR (dB) annotated bottom-right of each cell. At $\rho = 0.20$: trained `rich` ≈ 34 dB vs `block_dct_8` ≈ 28 dB (+6 dB) vs `dct` ≈ 22 dB (+12 dB).

6.2. Image 2 — fish silhouette

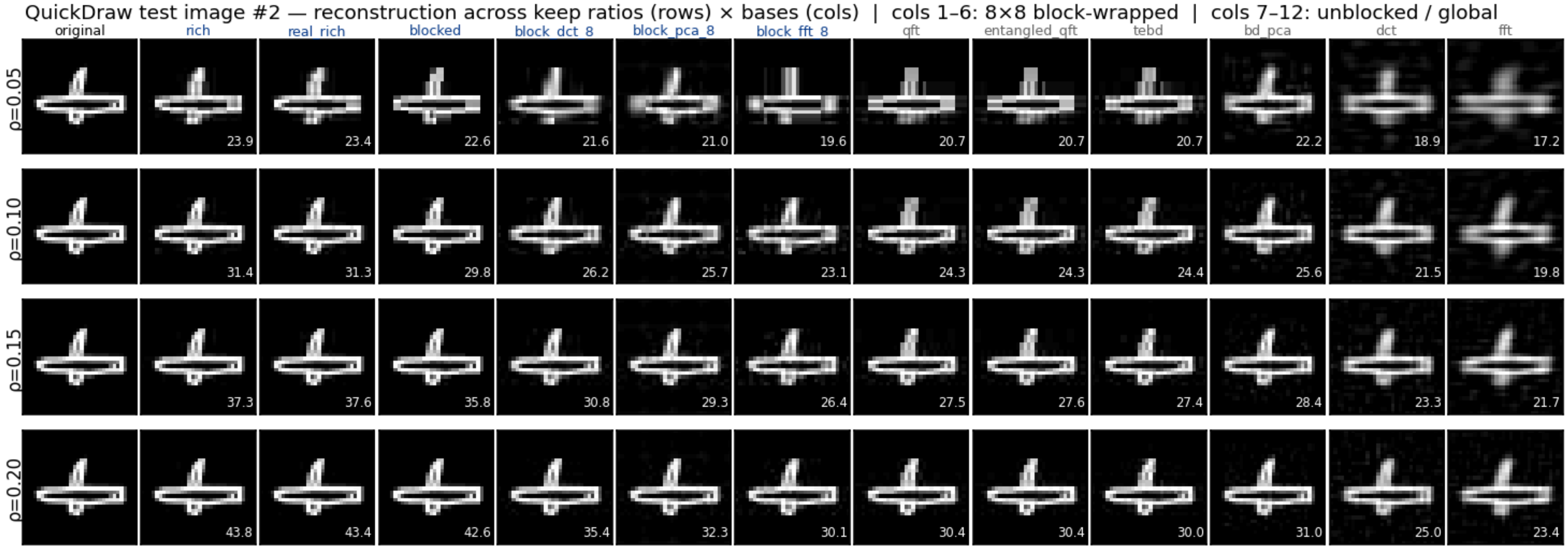


Figure 4: QuickDraw test image 2 reconstructed at the same four keep ratios. Same column layout as image 1 This image is among the strongest cases for trained tensor-network bases: at $\rho = 0.20$, trained `rich` ≈ 44 dB vs `block_dct_8` ≈ 32 dB (+12 dB) vs `dct` ≈ 25 dB (+19 dB). Even at $\rho = 0.05$ (top row) `rich` and `real_rich` already preserve the recognisable fish-stroke topology while `dct` and `fft` collapse to a blurred blob.

7. Matching summary — bench name → paper figure

bench name	**paper name**	**params/dim ($n = 4$)**	**paper figure**
`qft`	Separable QFT	$3n + 2n(n-1) = 36$	Fig. 1d, Tab. 1
`entangled_qft`	Entangled QFT	$36 + n = 40$	Fig. 3, Eq. (entangled_qft)
`tebd`	TEBD ring	$3n + 2n = 20$	App. C left, Fig. (tebd_circuit)
`mera`	MERA hierarchical	$3n + 4(n-1) = 24$	App. C right, Fig. (mera_circuit)
`blocked`	Blocked QFT	15 ($m_{\text{in}} = 3$)	§`block_wrapper`, Fig. (block_basis_circuits) bottom
`rich`	RichBasis	54 ($m_{\text{in}} = 3$)	Fig. (block_basis_circuits) middle
`real_rich`	RealRichBasis	21 ($m_{\text{in}} = 3$)	Fig. (block_basis_circuits) top — headline

Table 4: Mapping from `pdft_benchmarks` basis name to the paper's naming and circuit figure. Param counts use the conventions in main.tex §`sec:qft_basis` and §`sec:block_wrapper`.